L_1 and L_2 Regularization

Regularizers are based on:

Penalize large values of W (model parameters)

- This have the effect of better generalization of the model
- Since there is a contradiction between model training (using training dataset) and model generalization, the generalization term is added to the objective loss function with a customized weigh factor (do regularization to what extent)
- Penalization may be applied on the Squared Norm of the Weight Vector
  This is called L_2 regularization (also know as Ridge Regularization)
- Penalization may also be applied on the sum of Absolute values of the Weight Vector
  This is called L_1 regularization
- Regularization is particularly important for Small number of samples (N) and Large vector Length (d)